

FRAM

Forward. Automated disaster response planning.

The Problem

When disaster strikes, the data pipeline breaks down — costing lives.

01

Slow & Manual

NGOs and agencies like MapAction must manually label and digitise remote sensing imagery after every disaster — painstaking specialist work that takes days the world doesn't have.

02

Scarce Experts

The process depends entirely on skilled geospatial volunteers being available at exactly the right moment. That guarantee cannot be made when it matters most.

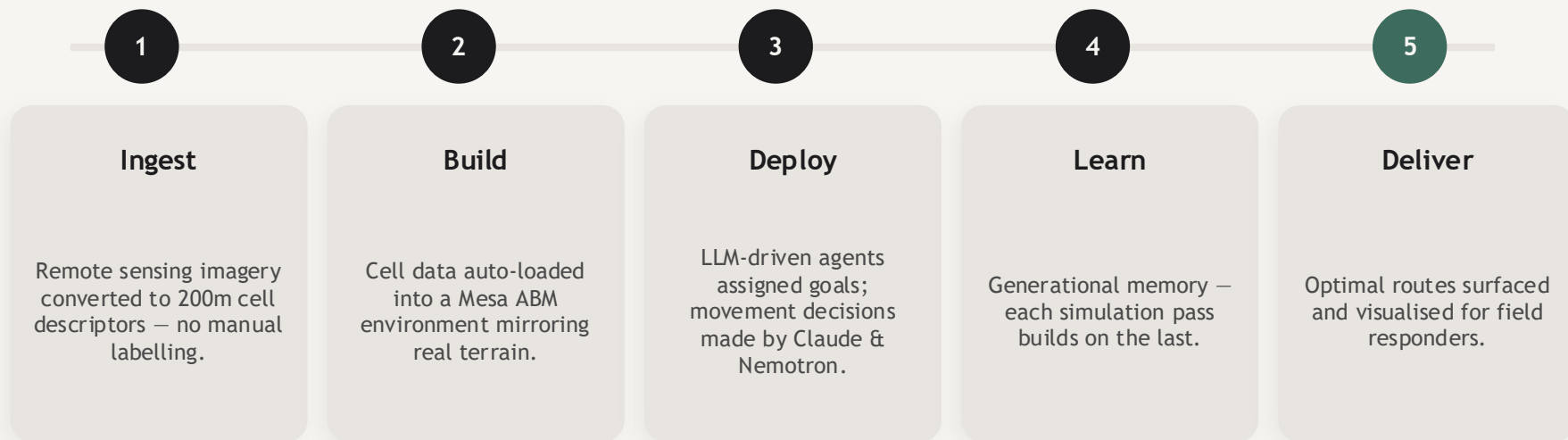
03

Data Gaps

Accurate, current geographic data is assumed to exist on demand. In reality, reliable ground truth is rarely available in the critical first hours following a disaster.

The Solution

A fully automated pipeline — from satellite imagery to field-ready routes.



Use case — flooding in Mozambique: planning evacuation & supply routes before field teams deploy.

How It Was Built

- 1 Remote sensing imagery processed into 200m cell descriptors
- 2 Cell data loaded into Mesa (Python ABM) to construct a digital world
- 3 Agents created with explicit goals: reach destination, carry supplies
- 4 LLMs make real-time movement decisions for each agent
- 5 Iterative generational runs – memories passed forward, strategy compounds
- 6 Best routes surfaced and visualised for real-world responders

Zero Labelling

Automation replaces days of manual specialist work

Generational Memory

Strategy compounds across simulations risk-free

LLM Reasoning

Agents reason and adapt – no hand-coded rules

Field-Ready Output

Plans delivered before boots hit the ground

Technology Stack

Code Generation

- Claude Code — Anthropic
- GitHub Copilot
- OpenAI Codex

ABM Framework

- NetLogo (initial mockup)
- Mesa — Python ABM SDK
- Python 3.x runtime

Agent Intelligence

- NVIDIA Nemotron LLM
- Doubleword API endpoints
- Generational memory layer

Geospatial Input

- Remote sensing imagery
- 200m cell descriptor format
- Automated digitisation pipeline

Our Partners

Built with world-class technology sponsors.



Nemotron LLM

Powers every agent decision and generational learning pass through Nemotron – an advanced LLM optimised for agentic reasoning and simulation workloads at scale.

Doubleword

API Endpoints

Provides the API infrastructure connecting the ABM simulation layer to real-time language model inference – enabling agent movement processing at scale.

ANTHROPIC

Claude Code

Claude Code accelerated the entire FRAM codebase – from ABM environment construction to agent logic – enabling a small team to move at unprecedented speed.

Move Forward.

FRAM automates the path from satellite data to life-saving field decisions.

NVIDIA Nemotron

Doubleword API

Anthropic Claude Code

FRAM · Framover · Forward